

Online Business and Information Technology - NotebookLM

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Discuss Online Business.

Online business involves using the online environment to market products, grow a business, and facilitate various types of transactions, often leveraging sophisticated information technology (IT) and information systems (IS). Doing business electronically requires firms and their employees to possess skills in online marketing techniques, technical support, and security knowledge to navigate threats.

Online Marketing and Advertising

The online environment helps businesses take their marketing messages to a national or global audience. Key strategies and tools used include:

- 1. Online Advertising:** This encompasses any form of advertising that uses the Internet to market messages to customers, such as banner ads, Facebook campaigns, spam emails, and Twitter messages. Sophisticated advertising methods, enabled by increased high-speed broadband and mobile device access, include using QR codes, animated ads via HTML5/Java, and iAds on mobile apps.
- 2. Pay-Per-Click (PPC) Advertising:** Advertisers pay only for the number of times a web surfer clicks on their ads. Programs like Google's AdWords successfully generate revenue this way [56, 143].
- 3. Viral Marketing:** This technique uses social networks, email, and websites to spread brand awareness. A viral campaign is often subtle, relying on markets), online advertising faces drawbacks. These include the **annoyance factor** of pop-up ads and interstitials, the fear of **spam**, and **click fraud**, where a competing company repeatedly clicks on a competitor's ads

Transaction Type	Description	Examples and Models
Business-to-Business (B2B)	Exchange of products, services, and information between businesses on the Internet.	Company websites (e.g., UPS site for business clients), e-procurement, procurement exchanges (e.g., Alibaba.com), specialized industry portals (niche portals), brokering sites, and information sites (infomediaries).
Business-to-Consumer (B2C)	E-commerce taking place directly between businesses and consumers.	Online intermediaries (e.g., Amazon.com), advertising-based models (high-traffic or niche), community-based models (e.g., Steam), and fee-based models (requiring subscription fees, e.g., Netflix). B2C businesses with both an online presence and physical stores are referred to as "click and mortar".
Consumer-to-Consumer (C2C)	Transactions between individual consumers,	Marketplaces with intermediaries (e.g., eBay or Etsy), peer-to-peer (P2P) file/music sharing, and social networking sites.

sometimes involving third parties.

Challenges of E-commerce

Online businesses face challenges related to taxation and jurisdiction:

- **Taxes:** Most Internet sales are tax-free, creating problems for states dependent on sales tax. If an e-business has a physical presence (store, office, or warehouse) in a state, that state may require it to impose a sales tax. Some states impose a "**use tax**," which consumers are required to pay on items purchased over the Internet if the seller did not collect state taxes.
- **Legal Jurisdiction:** Due to the Internet's broad geographic territory, defining legal jurisdiction for online-only businesses is difficult. Laws like the Digital Millennium Copyright Act (1998) must be followed to protect the rights of buyers and sellers and copyrighted material.

Technology, Information, and Business Intelligence

The backbone of online business is **Information Technology (IT)**, which includes the equipment and techniques used to manage and process information, and **Information Systems (IS)**, also called **Management Information Systems (MIS)**, which focus on applying IT to solve business and economic problems.

- **Data and Information:** Businesses collect **data**—raw, unorganized facts. **Information** is data that has been organized or arranged in a way that makes it useful for analysis, control, and decision-making. Databases and database management systems (DBMSs) collect, organize, store, and retrieve this data.
- **Data Management:** A **data warehouse** combines many databases across the whole company into one central database supporting management decision-making. **Data mining** involves sophisticated database applications that look for hidden patterns in data to help a business uncover relationships and patterns.
- **Big Data Analysis:** The use of very large data sets (like petabytes of transaction data) and special software tools to extract new trends and insights is called **Big Data analysis**. The collection of this vast consumer data raises significant privacy concerns, requiring companies to balance their information needs with protecting individual consumer rights.
- **Business Intelligence Software:** Management Support Systems (MSSs) use the company's data to perform high-level analyses that aid in better decision-making. Tools include:
 - **Decision Support Systems (DSS):** Software that helps managers make decisions using interactive computer models, often used for answering "what if" questions.
 - **Executive Information Systems (EIS):** Customized for individual executives, providing specific information for strategic decisions.
 - **Online Analysis Packages (OLAP):** Software designed to help people combine multiple pieces of information for tasks such as sales reporting, budgeting, and forecasting.

Security and Protection

Online businesses are highly vulnerable to security threats. Companies must develop specific security strategies and formal written information security policies to proactively prevent problems.

Major threats include:

- **Adware and Spyware:** Adware displays pop-up ads, while spyware tracks personal information and passes it to third parties, potentially leading to identity theft or fraud. Insidious spyware can capture keystrokes, stealing passwords and confidential information.
- **Phishing:** A common way to trick online users into sending private information (like credit card numbers or passwords) directly to hackers by imitating legitimate communications (e.g., from a bank or online retailer).
- **Hackers and Data Breaches:** Individuals who gain unauthorized entry into a computer system to disrupt operations or access protected data. Data breaches can result in millions of accounts being exposed, costing companies billions of dollars and damaging their reputations.

Impact of Social Media and Mobile Technology

New advances in social networking and mobile technologies significantly impact how businesses operate and interact with customers.

- **Social Networking:** Firms use social networks to communicate internally (**enterprise social networking**) to increase collaboration and speed up decision making, and externally to generate interest and sales.
- **Social Media Monitoring:** Companies use software (like Brandwatch or UberVU) to keep watch on external messages and information being spread about their products in cyberspace, identifying problems and opportunities.
- **Crowdsourcing and Crowdfunding:** **Crowdsourcing** uses suggestions from people on social media and the Internet to help a business make decisions (e.g., product development ideas). **Crowdsourced funding (crowdfunding)** leverages social networks and the Internet to raise capital for new ventures by having many people contribute small amounts of money.
- **Mobile Devices and Cloud Computing:** Businesses require a mobile component to support employees and serve customers, providing constant access to emails, calendars, and Internet resources. **Cloud computing** supports the mobile market by storing and retrieving information, resources, and software on remote servers accessed via the Internet. This allows for automatic backups and instant synchronization of data, increasing productivity and efficiency for mobile workers.

What is the CIO's role?

The **Chief Information Officer (CIO)** is an executive entrusted with the responsibility of managing a company's entire information resources. This position is typically at the same organizational level as the Chief Financial Officer (CFO).

The CIO oversees a firm's information processing, and their functions span several critical areas of technology and business strategy:

Management of IT and Systems

- **Overseeing Design and Development:** The CIO is in charge of overseeing how the firm's Information Technology (IT) system is designed and developed.
- **System Decisions:** The CIO decides whether a firm should update or replace its computer systems and software.
- **Hardware and Software Selection:** They are responsible for making decisions about what hardware and software the company purchases and when it should be updated.

- **Budget Management:** The CIO manages the budget, ensuring a balance between the benefits derived from technology and the rising costs associated with having the latest systems and upgrading them. This role requires evaluating whether the return on investment (ROI) justifies the cost of new systems.

Policy and Strategy Development

- **E-Business Opportunities:** CIOs are involved in creating business and e-business opportunities.
- **Governing Policies:** They define governing policies for the company in crucial areas such as privacy and security.
- **Custom Software Specifications:** The CIO may create specifications for required custom software programs.

Collaborative Functions

The CIO requires excellent technical, leadership, and organizational skills. The IT department, which the CIO oversees, is comprised of professionals responsible for implementing these strategies, including maintaining security, designing networks and databases, selecting software, providing employee training, and managing mobile computing use.

Name three types of networks.

Based on the sources, three types of computer networks are:

1. **Local Area Network (LAN):** This type of network allows people at one site to exchange data and share the use of hardware and software from a variety of computer manufacturers. A LAN serves computers that are located at a single site.
2. **Wide Area Network (WAN):** This network connects computers located at different sites using telecommunications media such as phone lines, satellites, and microwaves. Several forms of WANs, including intranets and virtual private networks (VPNs), utilize internet technology.
3. **Virtual Private Network (VPN):** VPNs connect two or more private networks (like LANs) over a public network, such as the internet. They incorporate strong security measures to ensure that only authorized users can access the network and its sensitive corporate information.

Other network-related systems mentioned in the sources include:

- **Intranets:** These are private corporate networks, similar to LANs, that use Internet technology to link employees across various locations and with different types of computers. Intranets are a form of WAN.
- **Enterprise Social Networking:** This involves applying social networking products (like Facebook) in a corporate setting to increase collaboration and speed up decision making within the organization.
- **Wireless Local Area Networks (Wireless LANs):** These are also listed as types of networks used by companies.

What is viral marketing?

Viral marketing is a marketing technique that utilizes **social networks, email, and websites to spread brand awareness**.

This practice is often employed by marketers to get people to notice their advertisements.

Key characteristics and effects of viral marketing include:

- **Method of Spread:** A viral campaign relies on the user to be an active participant in spreading a message, often leveraging the digital nature of Web 2.0. It is seen as a Web 2.0 version of old-fashioned word of mouth, being more consumer-engaged, interactive, and collaborative.
- **Subtlety:** Viral campaigns are generally **more subtle** than overt advertising campaigns that rely on traditional airwaves with commercials.
- **Target Audience:** engaged, interactive, and collaborative.
- **Subtlety:** Viral campaigns are generally **more subtle** than overt advertising campaigns that rely on traditional airwaves with commercials.
- **Target Audience:** This technique is designed to appeal to consumers who may be resistant to traditional advertising.
- **Customer Connection:** Viral marketing gives a company the chance to promote a product or brand while allowing potential consumers to feel a **personal connection** to the brand by becoming part of a communal experience.
- **Example in Practice:** Hollywood movie studios have been known to use viral advertising. For example, the marketing campaign for the film *District 9* involved public service announcements, signs reading "Bus bench for humans only," and a phone number and website address that encouraged interaction and generated interest by having people "play along with the premise". This dedication to creating an entertainment experience shifts the paradigm of advertising's capabilities.

For example, a start-up venture selling a specialty vest to new moms might employ viral marketing by placing videos and images in popular forums and websites targeting new mothers and those throwing baby showers to build interest in the new product.

What is information technology?

Information technology (IT) encompasses the **equipment and techniques used to manage and process information** [1) encompasses the **equipment and techniques used to manage and process information**.

Key aspects of Information Technology include:

- **Components and Scope:** IT includes the computer-based systems designed and implemented for handling information. These systems involve the collection of raw data and the subsequent processing of that data into usable information.
- **Organizational Necessity:** Information is central to all organizations; without data regarding processes, participants, orders, products, inventory, scheduling, shipping, customers, suppliers, and employees, a business cannot operate.

- **Business Transformation:** Businesses rely on IT for a wide range of functions, from managing daily operations to making high-level strategic decisions. **Harnessing the power of IT gives a company a significant competitive advantage.**
- **Management:** The **Chief Information Officer (CIO)** is the executive responsible for managing a company's entire information resources and overseeing the design and development of the firm's IT system. The **IT department** consists of professionals who manage everything from hardware and software to networking strategies, security, and employee training.
- **Relationship with Information Systems (IS):** While IT includes the equipment and techniques, an **Information System (IS)**, also called a **Management Information System (MIS)**, focuses on applying IT to solve business and economic problems. IS professionals act as a bridge between technical knowledge and its impact on the business.

What are data warehouses?

Data warehouses are large, central databases that are used to support management decision-making [7, warehouses are large, central databases that are used to support management decision-making.

Key details about data warehouses include:

- **Structure:** A data warehouse combines many databases from across the entire company into this one central database.
- **Purpose:** They support management decision-making by consolidating data.
- **Capacity:** Data warehouses are capable of holding vast amounts of transaction data, sometimes reaching petabytes in size (a single petabyte can hold more than 13 years' worth of high-definition television video).
- **Relationship to Data Marts:** Data warehouses may contain many data marts, which are specialized subsets of the data warehouse, each focusing on a single area of data and organized for quick analysis.
- **Distinction from Production Databases:** Data warehouses are often created as database systems that are separate from a company's production databases.

What is data mining?

Data mining is a process that uses **sophisticated database applications to look for hidden patterns in a group of data.**

This technique is used to help a business uncover relationships and patterns within its vast stores of information.

Key details about data mining:

- **Goal:** Data mining increases the potential for tracking and predicting people's daily activities.
- **Data Source:** The data used for mining can come from gigantic data sets, such as petabytes of transaction data, or subsets of a data warehouse called data marts. When data is extracted from websites and saved to a database, that data can then be analyzed using data mining to determine patterns.

- **Business Application:** Data mining can be used to inform business decisions. For instance, a supermarket might use data mining to determine if it is worthwhile to keep a slow-selling product in stock because the few people who buy it are the most profitable customers.
- **Privacy Concerns:** The use of data mining significantly increases concerns about personal privacy. The collection of vast amounts of consumer data puts pressure on companies to balance their need for information with protecting individual consumer rights.

A notable example of data mining, which sparked worldwide news, involved the political data mining firm Cambridge Analytica, which gained access to the data of over 87 million Facebook users. This firm mined the collected data to create psychographic profiles of voters, which were then used to target individuals with tailored political content. This data was "scraped" and aggregated from users' public profile information.

What is a transaction processing system?

A **transaction processing system (TPS)** is the fundamental starting point of a firm's integrated information system.

The primary function of a TPS is to **collect and organize raw data** pertaining to a firm's operations.

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Specifically, the TPS:

- **Receives Raw Data:** It takes in raw data from both internal and external sources.
- **Organizes Data:** It prepares and organizes this raw data for storage in a database. This database is similar to a microcomputer database but is vastly larger.
- **Processing Method:** It organizes and processes this data using either batch or online processing.

The resulting organized data is then used by other elements of the company's Management Information System (MIS). For instance, Management Support Systems (MSSs) rely on the internal master database, which contains the data collected by the TPS, to perform high-level analyses that aid managers in decision-making.

What does information technology include?

Information technology (IT) includes the **equipment and techniques used to manage and process information** [1,) includes the **equipment and techniques used to manage and process information**.

Key components and aspects associated with Information Technology include:

- **Equipment and Techniques:** IT encompasses the hardware and methods necessary for handling information.
- **Data Management:** IT systems are essential because information is at the heart of all organizations. Without information about organizational aspects—such as **orders, products, inventory, scheduling, shipping, customers, suppliers, and employees**—a business cannot operate.

- **Competitive Advantage:** Harnessing the power of information technology gives a company a **significant competitive advantage**.
 - **Information Systems:** The application of IT to solve business and economic problems is the focus of an **Information System (IS)**, also known as a **Management Information System (MIS)**. These systems collect and store key data and produce the information managers need for analysis, control, and decision-making.
 - **IT Management:** The design and implementation of computer-based information systems is the responsibility of the **Chief Information Officer (CIO)** and the **IT department**. The IT department professionals are responsible for everything from hardware components and software programs to **networking strategies**.
- The sources also note that trends in information technology include **digital forensics**, the **distributed workforce**, and **grid computing**.

What is raw, unorganized data called?

The sources state that **raw, unorganized facts** that can be moved and stored are called **data**.

Businesses collect a great deal of this **data** in their daily operations. This data can consist of numbers, words, images, or sounds. When **data**.

Businesses collect a great deal of this **data** in their daily operations. This data can consist of numbers, words, images, or sounds. When data has been organized or arranged in a way that makes it useful, it is then referred to as **information**.

A firm's integrated information system starts with its **transaction processing system (TPS)**, which receives and organizes this raw data from internal and external sources for storage in a database.

Who manages all information resources?

The executive entrusted with the responsibility of managing a company's entire information resources is the **chief information officer (CIO)**.

The CIO is typically an executive position at the same organizational level as the Chief Financial Officer (CFO).

The responsibilities of the CIO include overseeing a)**.

The CIO is typically an executive position at the same organizational level as the Chief Financial Officer (CFO).

The responsibilities of the CIO include overseeing a firm's information processing. Their management role covers determining how the firm's Information Technology (IT) system should be **designed and developed**, and deciding whether to **update or replace computer systems and software**. They also define governing policies for the company in areas such as **privacy and security**.

While the CIO is the top executive responsible for information resources, the actual work of implementing and maintaining these systems falls to the **IT department**, which manages everything from hardware and software to networking strategies.

What is a local area network?

A **local area network (LAN)** is one of the two basic types of computer networks, distinguished by area network (LAN)** is one of the two basic types of computer networks, distinguished by the area they cover.

A LAN allows people at **one site** to exchange data and share the use of hardware and software from a variety of computer manufacturers. In essence, a LAN serves computers that are located at a **single site**.

Companies use various types of networks, including LANs and wireless local area networks, to share data and expensive hardware to improve operating efficiency.

LANs are contrasted with **wide area networks (WANs)**, which connect computers located at different sites using telecommunications media such as phone lines, satellites, and microwaves.

Additionally, **virtual private networks (VPNs)** connect two or more private networks, such as LANs, over a public network like the internet, utilizing strong security measures.

Name three types of networks.

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2. **Wide Area Network (WAN):** A WAN connects computers located at **different sites** using telecommunications media such as phone lines, satellites, and microwaves. Several forms of WANs, including intranets, virtual private networks (VPN), and extranets, use internet technology.

3. **Virtual Private Network (VPN):** VPNs connect two or more **private networks** (such as LANs) over a **public network**, like the internet. They include **strong security measures** to ensure that only authorized users can access the network and its sensitive corporate information. VPNs offer a cost-effective, secure connection between remote locations by using public networks.

Additionally, the sources mention **wireless local area networks**. They also describe **intranets** as private corporate networks, which are a type of WAN that uses Internet technology to link employees across many locations.

What is a transaction processing system?

A **transaction processing system (TPS)** is the fundamental starting point of a firm's integrated information system.

The primary function of a TPS is to **collect and organize raw data** pertaining to a firm's operations.

Specifically, the TPS:

- **Receives Raw Data:** It takes in raw data from internal and external sources.
- **Organizes Data:** It organizes these raw data for storage in a database. This database is similar to a microcomputer database but is vastly larger.
- **Processing Method:** It organizes and processes this data using either **batch or online processing**.

The data collected by the TPS forms the basis of the company's internal master database. This organized data is then used by other elements of the company's Management Information System (MIS), such as **Management Support Systems (MSSs)**, which perform higher-level analyses that help managers make better decisions.

How many users did Cambridge Analytica expose?

The political data mining firm Cambridge Analytica gained access to data belonging to **over 87 million Facebook users** due to a personality quiz application.

Specifically:

- While only **270,000 users downloaded the quiz app**, the Facebook data mining scandal also impacted all million Facebook users** due to a personality quiz application.

Specifically:

- While only **270,000 users downloaded the quiz app**, the Facebook data mining scandal also impacted all the friends of those who took the quiz.
- Cambridge Analytica was able to "scrape" and aggregate data from the public profile information of over 87 million Facebook users to match to their contact details.

What does B2B stand for?

The acronym **B2B** stands for **business-to-business** [20, 62B** stands for **business-to-business**.

B2B refers to one of the most common modes of online business transactions, involving the **exchange of products, services, and information between businesses** on the Internet.

B2B websites and transactions can be classified as:

- **Company websites** that target other companies and their employees to sell goods or services to business clients (e.g., UPS has a B2B site for businesses seeking shipping solutions).

- **E-procurement** (electronic procurement), which refers to the online purchase and sale of products and services between businesses (e.g., companies bidding on state projects through e-procurement websites).
- **Procurement exchanges**, which are marketplaces where companies can buy and sell goods or services at lower prices.
- **Specialized industry portals** (also called niche or vertical portals) that act as online gateways to major sources of information, discussion forums, and product listings (e.g., CEOExpress.com).
- **Brokering sites**, which are third-party sites acting as liaisons between providers and potential buyers of goods or services.
- **Information sites** (or infomediaries), which provide businesses with information about a particular industry, helping to facilitate and originate B2B traffic.

B2B is one of the three common types of online transactions, alongside business-to-consumer (B2C) and consumer-to-consumer (C2C). Additionally, B2B is one of the forms that an **enterprise portal** (an internal website providing proprietary corporate information to a defined user group) can take.

What is a LAN?

A **local area network (LAN)** is one of the two basic types of computer networks, distinguished by the area network (LAN)** is one of the two basic types of computer networks, distinguished by the area they cover.

Key characteristics of a LAN:

- **Function and Location:** A LAN allows people at **one site** to exchange data and share the use of hardware and software from various computer manufacturers. In short, a LAN serves computers that are located at a **single site**.
- **Business Use:** Companies use LANs (and wireless local area networks) to share data and expensive hardware, which helps improve operating efficiency.
- **Contrast with Other Networks:** LANs are contrasted with **wide area networks (WANs)**, which connect computers at different sites using telecommunications media like phone lines, satellites, and microwaves. LANs can also be connected over public networks using **virtual private networks (VPNs)**. Furthermore, **intranets** are described as private corporate networks, similar to LANs.

Define the term "data."

The term **data** is defined in the sources as **raw, unorganized facts that can be moved and stored**.

Additional details about data include:

- **Content:** Data can consist of **numbers, words, images, or sounds**.
- **Collection:** Businesses collect a great deal of data in their daily operations. They try to use all available resources to gather raw data.

- **Role in Systems:** Raw data is the starting point for information systems. The **transaction processing system (TPS)** receives and organizes raw data from internal and external sources for storage in a database.

- **Transformation to Information:** When data has been **organized or arranged in a way that makes it useful**, it is then called **information**. The extraction of information from raw data is critical to the success of many business enterprises.

For example, raw data from a grocery store's sales log can be processed into information that helps the store's owner understand purchasing patterns.

Discuss Technology in Business.

The use of technology in business today is crucial, influencing everything from daily operations and strategic decision-making to commerce, connectivity, and organizational structure.

Here is a discussion of the role and scope of technology in business, drawing on information systems, security challenges, and emerging trends:

I. The Foundational Role of Information Technology (IT)

Information Technology (IT) includes the **equipment and techniques used to manage and process information**. Information is at the **heart of all organizations**. Without key information regarding **orders, products, inventory, scheduling, shipping, customers, suppliers, and employees**, a business cannot operate. Harnessing the power of IT provides a company with a **significant competitive advantage**.

The IT Organization

The management of a company's entire information resources falls to the **Chief Information Officer (CIO)**, typically an executive at the same organizational level as the Chief Financial Officer (CFO). The CIO's role includes:

- Overseeing a firm's information processing.
- Determining how the IT system should be **designed and developed**.
- Managing the budget and striving for the highest possible return on investment (ROI).
- Defining governing policies for the company in areas such as **privacy and security**.
- Deciding whether to **update or replace computer systems and software**.

The **IT department** consists of professionals responsible for the implementation and maintenance of this infrastructure, covering everything from **hardware components and software programs to networking strategies**.

II. Managing Information: Systems and Data

The application of IT to solve business and economic problems is the focus of an **Information System (IS)**, also known as a **Management Information System (MIS)**. These systems collect, store, and process facts into usable information.

Data and Information

Businesses collect **data**, which are defined as **raw, unorganized facts that can be moved and stored**. Data can consist of **numbers, words, images, or sounds**. When data has been **organized or arranged in a way that makes it useful**, it becomes **information**.

Core Information Systems

A firm's integrated information system begins with its **Transaction Processing System (TPS)**.

- **TPS Function:** The TPS receives and organizes raw data from internal and external sources for storage in a database.
- **Databases:** Data is stored in electronic filing systems called **databases**. Using **database management systems (DBMS)** software, managers can easily enter, store, organize, select, and retrieve this data.

Management Support Systems (MSS)

To help managers make better decisions, **Management Support Systems (MSSs)** use the internal master database to perform high-level analyses:

- **Data Warehouses and Data Mining:** A **data warehouse** combines many databases across the entire company into one central database. **Data mining** involves sophisticated database applications that look for **hidden patterns** in data to track and predict daily activities.
- **Decision Support System (DSS):** A DSS helps managers by using **interactive computer models** to ask "what if" questions about real-world processes.
- **Executive Information System (EIS):** Similar to a DSS, but **customized for an individual executive** for strategic decisions.
- **Big Data Analysis:** The availability of gigantic data sets has led to the use of **Big Data analysis** and special software tools to find new trends and generate useful information, or "**business intelligence**".

III. Connectivity: Networks and the Cloud

Companies use networks to deliver information to employees, suppliers, and customers. A **computer network** is a group of two or more computer systems linked together to share data and information.

- **Local Area Networks (LANs):** Serve computers located at a **single site**.
- **Wide Area Networks (WANs):** Connect computers located at **different sites** using telecommunications media such as phone lines, satellites, and microwaves.
- **Intranets:** Private **corporate WANs** that use Internet technology to link employees across many locations.
- **Virtual Private Networks (VPNs):** Connect private networks (like LANs) over a **public network** (such as the internet), employing **strong security measures**.
- **Wireless Technologies:** Used by businesses to improve communications with customers, suppliers, and employees.
- **Cloud Computing:** A major trend where information, resources, and software are stored and retrieved on **remote servers accessed via the Internet** instead of a local server. This creates a virtual computing environment that allows a company to access resources on an as-needed basis.

IV. Technology in Commerce and Marketing

The Internet and mobile technologies have revolutionized business transactions and marketing.

Types of Online Transactions

The three most common modes of online transactions are:

1. **Business-to-Business (B2B):** Involves the **exchange of products, services, and information between businesses** (e.g., e-procurement, specialized industry portals, brokering sites).
2. **Business-to-Consumer (B2C):** Refers to e-commerce that takes place **directly between businesses and consumers** (e.g., online intermediaries like Amazon.com).
3. **Consumer-to-Consumer (C2C):** Transactions that take place between **individual consumers** (e.g., eBay, peer-to-peer file sharing).

Marketing and Social Media

- **Online Advertising:** Includes forms such as banner ads and social media campaigns. **Pay-per-click (PPC) advertising** requires advertisers to pay only for the number of times a web surfer clicks on their ads, making them effective because they are associated with relevant keywords.
- **Viral Marketing:** The practice of using social networks, e-mail, and websites to **spread awareness of a particular brand**.
- **Social Media Monitoring:** Companies use software (like Brandwatch or UberVU) to **keep watch on external messages** being spread about their products in cyberspace.
- **Crowdsourcing:** Businesses use information solicited from people on social media to **help make decisions** or develop products.
- **Mobile Devices:** Businesses find it critical to have a mobile component to support employees (e.g., remote access, expense reports) and to attract customers (e.g., mobile apps for ordering and tracking rewards).

V. Security and Privacy Challenges

The increasing dependence on computers requires substantial security planning and proactive strategies.

Security Threats

Businesses must protect against costly computer crimes and security problems:

- **Hackers:** Individuals who gain unauthorized entry into a system to disrupt operations or steal protected data. The retailer Target had a data breach impacting **more than 70 million people**, resulting in the theft of customer names and credit card numbers.
- **Malware:** This includes **adware** (displays banner or pop-up ads) and **spyware** (tracks personal information and passes it to a third party, potentially capturing keystrokes and passwords).
- **Phishing:** A common way to **trick online users into sending their personal information** (like credit card numbers) to hackers by impersonating a legitimate business.

Privacy Concerns

The massive electronic storage of personal information poses a threat to individual privacy. **Data mining** increases the potential for tracking and predicting people's daily activities. **Data brokers** collect and sell billions of data points gathered from public information, loyalty programs, and social media to make inferences about consumers.

A major incident involved the political data mining firm Cambridge Analytica, which gained access to data belonging to **over 87 million Facebook users** via a personality quiz application. Companies face the challenge of finding a balance between collecting the necessary information and **protecting individual consumer rights**.

Workplace Monitoring

In the workplace, the expectation of privacy is different than at home. **Electronic monitoring** is commonly used and legal (if employees are informed) to track employees' keystrokes, emails, Internet browsing histories, and mobile communication, often to prevent theft or fraud.

VI. Emerging Trends

IT is a continually evolving field:

- **Knowledge Management (KM):** Focuses on researching, gathering, organizing, and **sharing an organization's collective knowledge** to improve productivity, foster innovation, and **gain competitive advantage**.
- **Digital Forensics:** A relatively new science used by corporations, government agencies, and attorneys to **obtain electronic evidence** (like web pages, pictures, or e-mails) from computers and corporate networks during investigations.
- **The Distributed Workforce:** Many knowledge workers now work remotely (from home or on the road) rather than from a permanent office space.
- **Grid Computing:** Allows users to work collaboratively on a **virtual supercomputer** by clustering and integrating computers and applications from multiple sources, enabling them to harness unused power in existing networks.

What is a data mart?

A **data mart** is defined as a **special subset of a data warehouse**.

Data marts deal with a **single area of data** and are organized specifically for **quick analysis**.

They are contained within a **data warehouse**, which itself combines many databases across the whole company into one central database that supports management decision-making. The data warehouse, in turn, can hold vast amounts of transaction data, sometimes measured in petabytes.

Name two basic network types.

The sources state that there are **two basic types of networks** distinguished by the area they cover:

1. **Local Area Network (LAN):** A LAN lets people at **one site** exchange data and share the use of hardware and software from a variety of computer manufacturers. A LAN serves computers that are located at a **single site**.
2. **Wide Area Network (WAN):** A WAN connects computers at **different sites** via telecommunications media such as phone lines, satellites, and microwaves.

Companies use these types of linked computers to share data and expensive hardware to improve operating efficiency. The sources also note that several forms of WANs—specifically **intranets**, **virtual private networks (VPN)**, and **extranets**—use internet technology.

What is data mining used for?

Data mining is a process that involves using **sophisticated database applications that look for hidden patterns in a group of data**.

Data mining is used for the following purposes:

- **Tracking and Predicting Daily Activities:** The process increases the potential for **tracking and predicting people's daily activities**.
- **Business Decision Making:** It is used to **uncover relationships and patterns in the data that can help a business** make decisions.
- **Retaining Profitable Customers:** For example, a supermarket might use data mining to analyze whether to keep a slow-selling product in stock. If the analysis reveals that the few people who buy the product are among the supermarket's **most profitable customers**, it is likely worthwhile to keep the product to **retain their business**.
- **Psychographic Profiling (Political Context):** Data mining has been used to create **personality profiles of voters** by aggregating data collected online. These profiles are then used to **target individuals with specifically tailored political content**. For instance, the political data mining firm Cambridge Analytica used data access related to a personality quiz application to create these profiles.

The challenge for companies when utilizing data mining is to **find a balance between collecting the information they need** while also **protecting individual consumer rights**.

What is Information Technology (IT)?

Based on the sources, **Information Technology (IT)** is defined as the **equipment and techniques used to manage and process information**.

Key aspects of Information Technology in a business context include:

- **Focus:** IT encompasses the **design and implementation of computer-based information systems**.
- **Organizational Necessity:** Information is considered to be at the **heart of all organizations**. Without information concerning the processes and participants within an organization—including **orders, products, inventory, scheduling, shipping, customers, suppliers, and employees**—a business cannot operate.
- **Competitive Advantage:** Harnessing the power of IT provides a company with a **significant competitive advantage**. shipping, customers, suppliers, and employees—a business cannot operate.

- **Competitive Advantage:** Harnessing the power of IT provides a company with a **significant competitive advantage**.

- **IT Department Responsibilities:** The IT department is comprised of professionals responsible for everything from **hardware components and software programs to networking strategies**. They manage the design of networks and databases, select hardware and software, provide training, manage security (including protection against viruses and system failures), and ensure the operation and currency of equipment like computers and printers.

- **Leadership:** The executive entrusted with managing all information resources and overseeing the firm's information processing is the **chief information officer (CIO)**.

The term IT is closely related to an **Information System (IS)**, also called a **Management Information System (MIS)**. The distinction is that an **MIS is focused on applying IT to solve business and economic problems**. MIS professionals bridge the gap between technical knowledge and its impact on the business.

What does a CIO oversee?

The **Chief Information Officer (CIO)** is an executive position, typically at the same level as the Chief Financial Officer (CFO), who is entrusted with the responsibility of managing a company's entire information resources.

The CIO is responsible for **overseeing a firm's information processing** and has numerous specific functions:

- **System Design and Development:** Determining how the firm's Information Technology (IT) system should be **designed and developed**.

- **Hardware and Software Decisions:** Deciding whether a firm should **update or replace its computer systems and software**. This includes making decisions about what **hardware and software** the company purchases and when it should be updated.

- **Budget Management:** Managing a **budget** that balances the benefits of technology against the ever-rising costs of upgrading to the latest systems. The CIO strives for the **highest possible return on investment (ROI)** from IT assets.

- **Policy Definition:** Defining **governing policies** for the company in areas such as **privacy and security**. This also includes creating policies regarding intellectual property.

- **E-Commerce and Websites:** Creating and maintaining an **Internet storefront** and driving traffic to the corporate website.

- **Program Specifications:** Creating **specifications for required programs**.

- **Custom Software:** Creating, testing, and maintaining custom software programs.

- **Security:** Maintaining **secure interactions** within the firm.

- **Business Opportunities:** Being involved in creating business and e-business opportunities.

In essence, the CIO oversees the strategic direction, implementation, budget, and policies for all technology used to manage and process information within the organization.

What are data warehouses?

A **data warehouse** is a **central database** that is created by combining **many databases across the whole company**.

The primary purpose of a data warehouse is to **support management decision-making**.

Further details include:

- **Capacity:** Data warehouses can hold vast amounts of transaction data, sometimes measured in petabytes. (A single petabyte can hold more than 13 years' worth of high-definition television video).
- **Subsets:** Data warehouses may contain smaller components called **data marts**, which are special subsets of the data warehouse that each deal with a single area of data and are organized for quick analysis.
- **Context:** Data warehouses are created when companies begin to store large amounts of data in database systems that are separate from their production databases.

The information collected and organized in the data warehouse is used by **Management Support Systems (MSSs)** to perform high-level analyses that help managers make better decisions.

What is the purpose of an EIS?

The purpose of an **Executive Information System (EIS)** is to **support management decision-making**, particularly for **strategic decisions**.

Key characteristics and functions of an EIS include:

- **Customization for Executives:** An EIS is similar to a Decision Support System (DSS), but it is **customized for an individual executive**.
- **Strategic Focus:** These systems provide **specific information for strategic decisions**.
- **Consolidation and Summarization:** An EIS can **consolidate and summarize the transactions within an organization** by using both **internal and external sources**.
- **Interface:** Usually, an EIS has a more **graphical interface** compared to a DSS.
- **Role in MIS:** The EIS is a component of the **Management Support Systems (MSS)**, which are part of a typical company's Management Information System (MIS) designed to help managers make better decisions.

In contrast, a standard DSS often uses spreadsheets and may only be able to show one department or product at a time.

What does IT include?

Information Technology (IT) includes the **equipment and techniques used to manage and process information**. **Information Technology (IT)** includes the **equipment and techniques used to manage and process information**.

In a business context, the scope of IT is broad and encompasses the following elements:

- **Equipment and Techniques:** IT involves the methods and tools used to manage and process information.
- **Organizational Heart:** Information is considered to be at the **heart of all organizations**. A business cannot operate without information regarding its processes and participants, including **orders, products, inventory, scheduling, shipping, customers, suppliers, and employees**.
- **Infrastructure Components:** The IT department manages the entire infrastructure, which consists of **hardware components and software programs to networking strategies**.
- **System Implementation:** IT specifically refers to the **design and implementation of computer-based information systems**.
- **Specific Departmental Responsibilities:** The IT department is responsible for keeping all of a firm's equipment—such as **computers, printers, scanners, and other equipment**—operational and current. It is also responsible for managing the use of **mobile computing** throughout a company.

Harnessing the power of IT is critical for a company to gain a **significant competitive advantage**.

What are the types of networks?

A **computer network** is defined as a group of two or more computer systems linked together by communications channels to share data and information.

The sources distinguish several types of networks based on the area they cover and the technology they utilize:

I. Basic Types (Distinguished by Area Covered)

The sources identify two basic types of networks based on the area they cover:

1. **Local Area Network (LAN):** A LAN serves computers that are located at a **single site**. It allows people at that site to exchange data and share the use of hardware and software from various computer manufacturers.
2. **Wide Area Network (WAN):** A WAN connects computers at **different sites** via telecommunications media, such as phone lines, satellites, and microwaves.

II. Specialized Network Forms (Often utilizing Internet Technology)

Several specialized forms of networks use internet technology, often operating as or within WANs:

- **Intranets:** These are internal **corporate-wide area networks** that use Internet technology to link a company's employees. They are described as private corporate networks, similar to LANs, but they are technically WANs because they link employees in many locations.
- **Extranets:** These are forms of WANs that use internet technology to provide access to authorized **customers or vendors outside the company**. This contrasts with an intranet, which links only a company's employees.

- **Virtual Private Networks (VPNs):** VPNs are used by many companies to connect two or more private networks (such as LANs) over a **public network**, like the internet. VPNs include strong security measures to ensure only authorized users can access sensitive corporate information. VPNs are particularly useful for salespeople and telecommuters who need to access the company's network remotely.
- **Wireless Technologies:** Businesses use wireless technologies (including **wireless local area networks**) to improve communications with customers, suppliers, and employees. Examples of devices relying on wireless technology include cellular phones, mobile devices, and wireless headphones.
- **Enterprise Portals:** This is one type of intranet, specifically defined as an internal website that provides **proprietary corporate information to a defined user group**.

What is an enterprise portal?

An **enterprise portal** is defined as an **internal website** that provides **proprietary corporate information** is defined as an **internal website** that provides **proprietary corporate information** to a **defined user group**.

An enterprise portal is considered one **type of intranet**. Since intranets are internal corporate-wide area networks that use Internet technology to link employees, an enterprise portal operates within this structure.

Enterprise portals can take one of three forms:

1. **Business to Employee (B2E)**
2. **Business to Business (B2B)**
3. **Business to Consumer (B2C)**

Enterprise portals are described as opening the door to **productivity**.

What are data marts?

A **data mart** is defined as a **special subset of a data warehouse**.

Key characteristics of data marts include:

- **Location and Composition:** Data warehouses may contain **many data marts**.
- **Data Focus:** Data marts are designed to deal with a **single area of data**.
- **Organization:** They are specifically **organized for quick analysis**.

In the context of information systems, a **data warehouse** combines many databases across the whole company into one central database that supports management decision-making. Data marts are essentially segmented portions of this warehouse, used to isolate a single product or department.

What is data mining?

Based on the sources, **data mining** is defined as the process of using **sophisticated database applications that look for hidden patterns in a group of data**.

The primary use and implications of data mining include:

- **Tracking and Prediction:** Data mining increases the potential for **** applications that look for hidden patterns in a group of data****.

The primary use and implications of data mining include:

- **Tracking and Prediction:** Data mining increases the potential for **tracking and predicting people's daily activities**.

- **Business Intelligence:** It is used to **uncover relationships and patterns in the data that can help a business make decisions**. For instance, a supermarket might use data mining to decide whether to keep a slow-selling product in stock; if the few customers who buy it are among the company's most profitable, it would be worthwhile to keep the product to retain their business.

- **Privacy Concerns:** The practice of data mining contributes to the threat to personal privacy posed by huge electronic file cabinets full of personal information.

- **Targeted Profiling:** Data mining has been used to create **personality profiles of voters** by aggregating data collected online, which are then used to **target individuals with specifically tailored political content**. For example, the firm Cambridge Analytica gained access to data belonging to over 87 million Facebook users to create these psychographic profiles.

- **Data Scraping:** Data mining often operates on data that has been "scraped," which means amounts of data are extracted from websites and saved to a database where patterns can then be determined.

Companies face the challenge of finding a **balance between collecting the information they need** while simultaneously **protecting individual consumer rights**.

Discuss Security Challenges.

The increase in technology and dependence on computer systems present significant **security challenges** for businesses, in technology and dependence on computer systems present significant **security challenges** for businesses, encompassing everything from unauthorized access and cybercrime to the protection of private information and system stability.

The security challenges can be grouped into several major categories:

I. Cybercrime and Malicious Software

Firms must take steps to prevent costly computer crimes and problems, which include the following threats:

- **Unauthorized Access and Security Breaches:** Hackers, who are individuals gaining unauthorized entry into a computer system, may seek to disrupt operations or steal protected data. Companies like Orbitz, eBay, Home Depot, Target, Sony's PlayStation Network, TJ Max, and Under Armour have all experienced recent data breaches, resulting in billions of accounts being exposed. In 2017 alone, there were 16.7 million victims of identity fraud, with over \$16.8 billion stolen.

- One of the largest data breaches occurred when a Department of Veterans Affairs (VA) employee violated policy by taking home a laptop, which was later stolen, containing the names and Social Security numbers of over 26 million veterans.

- Target experienced a breach that impacted more than 70 million people, with stolen information including customer names, credit card numbers, and email addresses, costing the company nearly half a billion dollars.

- Sony Pictures Entertainment was severely compromised when hacker software stole and erased data from thousands of computer systems, making unfinished movie scripts, Social Security numbers, and salary information public.

- **Computer Viruses, Worms, and Trojan Horses:** These malicious programs are a major category of security problems.

- **Adware and Spyware:**

- **Adware** is software that displays banner ads or pop-up ads while a program is running, which can be annoying.

- **Spyware** is software that tracks personal information and passes it on to a third party without the user's knowledge, operating in the Internet connection's backchannel. The most insidious spyware can capture keystrokes, stealing passwords and confidential information, which grants hackers unlimited access to company networks and can cripple businesses.

- **Phishing:** This is a common method used by hackers to trick online users into sending their personal information, such as credit card numbers, directly to them. Phishing emails impersonate legitimate businesses, often using official logos, and link to fake websites designed to look like authentic forms to capture sensitive data.

- **Spam:** Unwanted electronic junk mail is listed as a major category of computer crime.

- **Software and Media Piracy:** Illegally copying or distributing software and media is another security and legal challenge.

- **Deliberate Damage:** This includes physical or electronic damage to equipment or information.

II. Privacy and Data Usage Concerns

The vast amounts of data collected and stored by businesses pose a fundamental threat to personal privacy.

- **Data Mining and Profiling:** The process of **data mining** (looking for hidden patterns in data) increases the potential for **tracking and predicting people's daily activities**. Companies collect and sell information from various sources like warranty cards, credit card records, website registrations, and loyalty programs. This practice is used, for example, to create psychographic profiles of voters for targeted political content. Companies face the challenge of balancing the need to collect information with the necessity of **protecting individual consumer rights**.

- **Big Data Analysis:** The availability of gigantic data sets leads to new value in collecting billions of clicks, purchases, and behaviors. Data brokers gather and sell this information, sometimes combining pieces of public and

marketing data to make inferences about individuals (e.g., liking sports or having children). There is growing concern about how to protect data like private medical status or credit standing.

- **Employee Privacy and Electronic Monitoring:** In the workplace, employee privacy expectations are different than at home. Companies can legally monitor employee actions on employer-provided computer equipment. **Electronic monitoring** is commonly used to track keystrokes, emails, Internet browsing histories, and mobile communications to prevent theft, fraud, and loafing, and to improve security. However, extensive monitoring can increase employee stress and anxiety, potentially leading to health problems and job dissatisfaction.

III. Mitigating Security Challenges

To address these threats, companies need a proactive approach:

- **Security Strategies and Policies:** Companies must develop specific security strategies to prevent problems before they start. The first step is creating **formal written information security policies** to set standards and provide a basis for enforcement. These security plans require the support of top management.

- **Technology and Training:** Companies increase spending on technology to protect their IT infrastructure and data, utilizing tools such as **virus protection, firewalls, and employee training** in proper security procedures.

- **Disaster Preparedness:** Given the ever-increasing dependence on computers, plans must cover human error, power outages, equipment failure, terrorist attacks, and natural disasters.

- **VPN Security: Virtual Private Networks (VPNs)**, used to connect private networks over public ones like the internet, include **strong security measures** to ensure that only authorized users can access sensitive corporate information.

A lapse in corporate security, like a major data breach, can affect consumers by exposing their personal and identity information, raising questions about whether they should continue to purchase from certain companies.

The security challenges for businesses act like a perennial game of "cat and mouse," where businesses invest in defensive "fortresses" (like firewalls and written policies) while hackers constantly develop new ways (like phishing and spyware) to bypass those defenses and access the valuable information stored inside.

What is a management information system?

A **Management Information System (MIS)** is a comprehensive term for the **methods and equipment that provide information about all aspects of a firm's operations**. It is also referred to as an **Information System (IS)**.

The primary role of a Management Information System is to provide **managers with the information they need to make decisions**. MIS professionals **bridge the gap between purely technical knowledge and how it will impact a business**.

A typical company's Management Information System consists of several components:

1. **Transaction Processing System (TPS):** This system **collects and organizes operational data** on the firm's activities. The TPS receives raw data from internal and external sources and prepares this data for storage in a database.

2. **Management Support Systems (MSSs):** These systems use the internal master database to perform **high-level analyses that help managers make better decisions**. MSSs include:

- **Information-Reporting System:** Provides information based on the data collected by the TPS to the managers who require it.
- **Decision Support Systems (DSS):** Uses computer models to **assist in answering "what if" types of questions**.
- **Executive Information Systems (EIS):** These are systems **customized to the needs of top management** and provide specific information for strategic decisions.
- **Expert Systems:** Gives managers **advice similar to what they would get from a human consultant**.

3. **Office Automation System:** This is also a component of a typical company's MIS.

In essence, an MIS focuses on **applying Information Technology (IT) to solve business and economic problems**. It is responsible for collecting and storing a company's key data and producing the usable information necessary for **analysis, control, and decision-making**.

What is a transaction processing system?

A **Transaction Processing System (TPS)** forms the starting point of a firm's integrated information system. It is a foundational component of a company's overall Management Information System (MIS).

The main functions of a TPS are:

- **Data Collection and Organization:** The TPS is responsible for **collecting and organizing operational data** on the firm's activities.
- **Data Input:** It **receives raw data from internal and external sources**.
- **Data Preparation:** It **prepares these raw data for storage in a database**.
- **Processing Methods:** Data is received and organized for storage using either **batch or online processing**.

The data collected by the TPS is essential because it feeds into other parts of the MIS, such as the **Management Support Systems (MSSs)**, which then use this data to perform higher-level analyses to help managers make better decisions.

What is the purpose of data?

The purpose of **data** is to serve as the **raw, unorganized facts that can be moved and stored**.

Key purposes and roles of data are:

- **Foundation for Information:** Data consists of representations of a fact or idea, which can include **numbers, words, images, or sounds**. The crucial step is that raw data must be **organized or arranged in a way that makes**

the data useful to become **information**. The **extraction of information from raw data** is critical to business success.

- **Organizational Operations:** Businesses collect a great deal of data in their daily operations. Information (which is processed data) is considered to be at the heart of all organizations, and without it—including data concerning **orders, products, inventory, scheduling, shipping, customers, suppliers, and employees**—a business cannot operate.

- **Input for Systems:** Data serves as the input for systems like the **Transaction Processing System (TPS)**, which **receives raw data from internal and external sources** and prepares this data for storage in a database.

- **Support for Decision-Making:** Data is collected and processed into usable information that managers need for **analysis, control, and decision-making**. This is achieved through systems like **databases**, which are electronic filing systems designed to **collect and organize data and information**.

- **Business Intelligence:** Data can be analyzed using specialized software systems (such as Data Mining and Big Data analysis) to **uncover relationships and patterns** that help a business make better decisions, a process often referred to as transforming data into **useful business intelligence**.

In essence, data is the unprocessed material that, once organized and analyzed, becomes the vital information and intelligence necessary for a firm's operations and strategic decision-making.

Discuss Social Media Impact.

The rise of social networking and mobile technologies has significantly **impacted business operations**. These advances are changing the face of business in unexpected ways, creating new employment and marketing opportunities, but also introducing new dangers.

The impact of social media can be discussed across several key business areas:

1. Marketing and External Communication

Social media is now a critical tool for reaching customers and advertising products.

- **Online Advertising:** Online advertising now includes Facebook campaigns and messages sent on Twitter.

- **Viral Marketing:** Marketers use social networks, email, and websites for **viral marketing**, which spreads brand awareness. Viral marketing is considered a Web 2.0 version of old-fashioned word of mouth, relying on the user to be an active participant in spreading the message. This technique allows potential consumers to feel a personal connection to the brand by becoming part of a communal experience.

- **Generating Revenue:** Firms use social networks to generate interest in their products and increase revenues. For example, the @DellOutlet Twitter account has generated millions of dollars of sales for Dell.

- **Brand Perception and Risk:** Social media can also pose a threat, as negative comments posted on blogs or Facebook can have a real negative impact. One upset customer starting to drive away others is a known risk. Furthermore, careless or "tasteless" social media posts from companies can lead to public backlash.

2. Internal Operations and Business Intelligence

Companies utilize social media tools both internally and as sources of external information for decision-making.

- **Enterprise Social Networking:** Companies are applying social network tools in a corporate setting through **enterprise social networking**. Tools like enterprise versions of Facebook and Twitter, and workplace wikis such as Yammer, are used to increase collaboration, speed up decision making, and improve customer experience.
- **Social Media Monitoring (SMM):** It is critical for firms to monitor what is being said about their products in cyberspace. This practice, called **social media monitoring**, involves using software (like Brandwatch and UberVU) to collect, analyze, and report on patterns and trends in social media mentions.
 - Dell, for instance, established a Social Media Listening Command Center, where employees monitor over 25,000 web posts daily.
 - SMM is also used for technical support, matching social media comments with specific customer records to respond quickly to problems.
- **New Jobs:** The rise of social media has led to the creation of roles such as **social media managers** or social media marketing specialists. These professionals monitor networks, generate "buzz," listen for problems, and manage engagement with customers.

3. Crowdsourcing and Funding

Social networks provide new mechanisms for gathering input and capital.

- **Crowdsourcing:** Businesses are using information solicited from people on social media and the Internet to help make decisions, a process called **crowdsourcing**. For example, Dell uses its IdeaStorm website to allow customers to post and promote ideas for product development. ModCloth uses a "Be the Buyer" program, encouraging customers to vote online on clothing styles to influence buying decisions.
- **Crowdsourced Funding (Crowdfunding):** Social networking is a mechanism for funding start-ups. **Crowdsourced funding** allows a great number of people who believe in a company or product to contribute small amounts of money to help it get started.

4. Privacy Concerns

The data generated by social media activity is valuable but raises privacy issues.

- **Data Aggregation and Profiling:** The availability of gigantic data sets means there is new value in collecting the billions of clicks, purchases, and behaviors of consumers. Data brokers gather and sell this information, including tracking usage of social media. Software systems combine data to make inferences about individuals, such as whether they like sports or have children.
- **Data Mining Scandals:** Sophisticated database applications, through **data mining**, increase the potential for tracking and predicting daily activities. The sources highlight the controversy involving Cambridge Analytica, which gained access to data from over 87 million Facebook users via a personality quiz application to create psychographic profiles of voters for targeted political content. Following this disclosure, Facebook implemented numerous changes regarding data access.

Discuss Mobile Technology.

Mobile technology, including cellular phones, tablets, and other mobile devices, has had a significant impact on modern business operations, influencing both internal processes and external customer engagement.

Here is a discussion of the impact and use of mobile technology:

I. Definition and Ubiquity of Mobile Technology

Wireless technology is commonplace today and includes devices such as **cellular phones, mobile devices, garage door openers, and television remote controls**. Businesses utilize wireless technologies to **improve communications with customers, suppliers, and employees**. The increasing number of mobile devices has supported a large increase in the growth of mobile technology.

II. Business Implementation and Strategy

Firms find it critical to have a **mobile component** to both **support their employees** and to **attract and serve customers**.

- **Employee Support:** A firm's sales staff requires constant access to **e-mail, calendar and contacts, and Internet resources**. Mobile devices allow employees to make **video phone calls** with applications like Facetime, **see data mapped to their current locations**, and **create and send photos and videos**. The IT department is specifically responsible for **maintaining and managing the use of mobile computing** throughout a company.
- **Customer Service:** Customers demand ways to **find the closest locations at which to shop** and to **place orders from their smartphones**. For example, the airline Go! was forced to double the number of \$1 tickets it offered to win over customers when its promotion crashed its website, highlighting the critical nature of reliable customer access.
- **Mobile Apps and Functionality:** There are over **2 million apps** available for mobile devices. For business, many useful apps focus on **communications, task management, and managing data**.
 - Communication apps allow users to conduct **WebEx conferences**, integrating slide shows, file transfer, and multiple voice and audio feeds.
 - Task management apps like Omnifocus organize schedules beyond a simple calendar, displaying tasks that require nearby resources or those on hold.
 - Sales force apps organize **new leads and customer requests**, manage **salespeople's expense reports**, or **process invoices** right from their tablets or phones.

III. Strategic Planning for Mobile Devices

When a firm selects which mobile devices to purchase, it must consider several issues:

- **Device Selection:** Determining what kind of mobile devices (e.g., phones, tablets) and which brand to use.
- **User Deployment:** Deciding which employees should use them, and what kind of devices customers are using.
- **Policy:** Determining if employees should be able to use their personal mobile phones for work.
- **Integration and Security:** Assessing how easily new devices integrate with existing mail systems, what kind of **encryption and data protection** is available, and whether necessary applications exist or can be developed **in-house**.
- **Platform Support:** Externally, a company ideally provides mobile solutions for various platforms (iPhone, Android, cellular phones). However, the cost of developing and maintaining multiple applications means a business must understand its client base before deploying mobile products.

IV. Technology Trends Supporting Mobile Use

Mobile technology growth is heavily supported by **cloud computing**.

- **Cloud Computing:** This practice involves storing and retrieving information, resources, and software on **remote servers accessed via the Internet**. Cloud computing makes a business **free from worries that data could be lost** and increases employee productivity through the **instant synchronization of data**. For instance, Apple's iCloud service instantly makes a photograph taken on an iPhone available on an iPad or home PC.

V. Mobile-Related Marketing

Mobile technology facilitates modern marketing and advertising techniques:

- **QR Codes:** These codes can be scanned with mobile devices, taking consumers from physical magazine ads or counter displays directly to a company's web content.
- **Sophisticated Advertising:** The increase in households with mobile devices has led to more sophisticated online advertising. Apple's iPhone apps can use a feature called **iAds**, which interact with the user within the original app.
- **Mobile Analytics:** Companies can use mobile analytics to display and analyze site traffic quickly when working with dealers in the field, pointing to specific customer behaviors that support stocking certain items.